Physical Chemistry CUME, March 8, 2019

Phase Equilibria

1. Derive a relation that describes how the chemical potential of pure water at a constant pressure changes by temperature, assuming no phase change occurs. [10 points]
2. Derive a relation that describes how the chemical potential of pure water at a constant temperature changes by pressure, assuming no phase change occurs. [10 points]
3. Find the boiling temperature of water at $p = 0.500\ atm$. First, derive a relation based on the Clapeyron equation below that allows you to calculate the boiling point of water at a given pressure. What assumptions do you need to make to derive this relation from the Clapeyron equation? [40 points]
4. Schematically draw the chemical potential of pure water as a function of temperature (for T between $-20\ ^\circ C$ and $120\ ^\circ C$) at $p = 1\ atm$. [20 points]
5. Repeat Problem 4 for $p = 0.5\ atm$. Draw both curves schematically in the same plot. [20 points]

Useful Information

$$dG = Vdp - SdT$$

Clapeyron Equation: $\frac{dp}{dT} = \frac{\Delta_{tr}H}{T\Delta_{tr}V}$

Gas constant R = 8.314 J mol^{-1} K^{-1}

$\Delta_{vap}H(water) = 40.66$ kJ at $100\ ^\circ C$